TD n° 1.

Ex 1. Rocq

Ex 2. Rocq

Ex 3. If $\Gamma \vdash A$ is provable in NJ, take some proof tree in NJ for it, it is a valid proof tree in NJ + R, thus $\Gamma \vdash A$ is provable in NJ + R.

Conversely, take a proof tree π of $\Gamma \vdash A$ in NJ + R, by admissibility of R, every occurence of R in π can be replaced by part of a tree that doesn't use R. Doing this starting from the leaves yield a valid proof tree for $\Gamma \vdash A$ in NJ.

Ex 4.

1. Take a proof tree of $\Gamma, A, B \vdash C$. Replace each occurence of

$$\frac{}{\Gamma, A, B, \Delta \vdash A}\,Ax \qquad \text{with} \qquad \frac{\dfrac{}{\Gamma, A\wedge B, \Delta \vdash A\wedge B}\,Ax}{\Gamma, A\wedge B, \Delta \vdash A}\,\wedge eg$$

and

$$\frac{}{\Gamma, A, B, \Delta \vdash B}\,Ax \qquad \text{with} \qquad \frac{\dfrac{}{\tilde{\Delta} \vdash A\wedge B}\,Ax}{\tilde{\Delta} \vdash B}\,\wedge ed,$$

also replacing every sequent $\Gamma, A, B, \Delta \vdash D$ with $\Gamma, A\wedge B, \Delta \vdash D$.

2. By induction on the size of Γ, we apply Q1.

3. No, derivable rules cannot remove hypotheses (with the rules in NJ.

Ex 5.

1. $\dfrac{\Xi, \Gamma \vdash A}{\Xi, \Delta \vdash B}$ admissible iff $(\Xi, \Gamma \vdash A$ implies $\Xi, \Delta \vdash B)$ Ex 4.

iff $(\Xi \vdash \bigwedge\Gamma \Rightarrow A$ implies $\Xi \vdash \bigwedge\Delta \Rightarrow B)$

iff $\Xi \vdash (\bigwedge\Gamma \Rightarrow A) \Rightarrow (\bigwedge\Delta \Rightarrow B)$ currying

iff $\Xi \vdash (\bigwedge\Gamma \Rightarrow A \wedge \bigwedge\Delta \Rightarrow B$

2. Similarly to Q1, we know

$$\Xi, \Gamma \vdash (\bigwedge\Delta \Rightarrow A) \Rightarrow B$$

is provable, thus, by repeated currying, writing $\Delta = F_1, \dots, F_n$, the sequent

$$\Xi, \Gamma \vdash \underbrace{(F_1 \Rightarrow \cdots \Rightarrow F_n \Rightarrow A)}_{\Phi} \Rightarrow B$$

is provable; take π a proof tree for it.

$$\dfrac{\dfrac{\pi}{\Xi, \Gamma \vdash \Phi \Rightarrow B} \qquad \dfrac{\dfrac{\text{assumption}}{\Xi, \Gamma, \Delta \vdash A}}{\Xi, \Gamma \vdash F_1 \Rightarrow \cdots \Rightarrow F_n \Rightarrow A}\ \text{repeated } \Rightarrow_I}{\Xi, \Gamma \vdash B}\ \Rightarrow_E$$

Ex 6.

1.
$$\dfrac{\dfrac{\dfrac{\pi_1}{\Gamma \vdash A}}{\Gamma, \Delta \vdash A}\ wk \qquad \dfrac{\dfrac{\pi_2}{\Delta \vdash B}}{\Gamma, \Delta \vdash B}\ wk}{\Gamma, \Delta \vdash A \wedge B}\ \wedge I$$

2.
$$\dfrac{\dfrac{\Gamma \vdash \neg A}{\Gamma, \Delta \vdash \neg A}\ wk \qquad \dfrac{\Delta \vdash A}{\Gamma, \Delta \vdash A}\ wk}{\Gamma, \Delta \vdash \bot}\ \neg E$$

$$\dfrac{\dfrac{\Gamma \vdash A \vee B}{\Gamma, \Delta, \Sigma \vdash A \vee B}\ wk \qquad \dfrac{\Delta, A \vdash C}{\Gamma, \Delta, \Sigma, A \vdash C}\ wk \qquad \dfrac{\Sigma, B \vdash C}{\Gamma, \Delta, \Sigma, B \vdash C}\ wk}{\Gamma, \Delta, \Sigma \vdash C}\ \vee E$$

$$\dfrac{\dfrac{\Gamma \vdash A \Rightarrow B}{\Gamma, \Sigma \vdash A \Rightarrow B}\,\text{wk} \qquad \dfrac{\Delta \vdash B}{\Gamma, \Delta \vdash B}\,\text{wk}}{\Gamma, \Sigma \vdash B}\,\Rightarrow E.$$

Ex 7.

Define $C_\top := \bot \Rightarrow \bot$ and $C_{\neg A} := A \Rightarrow \bot$.

$$\dfrac{\dfrac{}{\Gamma, \bot \vdash \bot}\,\text{Ax}}{\Gamma \vdash C_\top}\,\Rightarrow I \quad \text{"}\top I\text{"}$$

$$\dfrac{\Gamma \vdash C_{\neg A} \qquad \Gamma \vdash A}{\Gamma \vdash \bot}\,\Rightarrow E \quad \neg E.$$

$$\dfrac{\Gamma, A \vdash \bot}{\Gamma \vdash C_{\neg A}}\,\Rightarrow I \quad \neg I$$

Ex 8. Rocq

Ex 9. Rocq

$\neg_R A$ is equivalent to $\neg A$ iff $\vdash \neg R$ is provable.

TD n° 2.

I) Preliminaries

Exercise 1.

Take R a relation over λ-terms.
Then for all $M \to_R N$, for all $C[\,]$, we have $C[M] \to_R C[N]$

by def. of $\to_R$.

Exercise 2

By induction on $M R^* N$,

if $M \; R^0 \; N$, i.e. $M = N$, then $C[M] \; R^0 \; C[M] = C[N]$.

if $M \; R \; N \; R^* \; P$ then by ih $C[N] \; R^* \; C[P]$

and $C[M] \; R \; C[N]$ by congruence.

II) The complexity of the simply-typed β-reduction.

Exercise 3.

For every $n \in \mathbb{N}$,

$$D_{A_n} = \lambda f_{n+1} : A_n \to A_n . \lambda f_n : A_n, \underbrace{f_{n+1} \underbrace{(f_{n+1} \; f_n)}_{: A_n}}_{: A_n} : \underbrace{A_{n+1} \to A_{n+1}}_{= A_{n+2}}$$

By induction on $k \in \mathbb{N}$, we show

$$T_{A_n, k} : A_{n+2} \to A_{n+3} \to \dots \to A_{n+2+k}$$

$$T_{A_n, 0} = D_{A_n} : A_{n+2}$$

$$T_{A_n, k+1} = T_{A_{n+1}, k} \; D_{A_n}$$

$$: A_{n+2}$$

Exercise 4.

By induction on k,

- $T_{A_n,0} = D_{A_n}$ is already in β-normal form

$$\lambda f_{n+1}.\lambda f_n.\ f_{n+1}^{2}\ f_n.$$

as $\sigma(1)=2$

- $T_{A_n,k+1} = T_{A_{n+1},k}\, D_{A_n} \longrightarrow_\beta^* (\lambda f_{n+1}.\lambda f_{n+1}\, f_{n+2}^{\sigma(k+1)} f_{n+1})$

$$(\lambda f_{n+1}.\lambda f_n.\ f_{n+1}^{2}\ f_n)$$

$$\longrightarrow_\beta \lambda f_{n+1}(\lambda f_{n+1}.\lambda f_n.\ f_{n+1}^{2}\ f_n)^{\sigma(k+1)} f_{n+1}$$

$$\longrightarrow_\beta^* \lambda f_{n+1}.\lambda f_n.\ f_{n+1}^{2^{\sigma(k+1)}}\ f_n.$$

III | Taming β-reduction using the size of types.

Exercise 5. By induction on M.

$$x \longrightarrow_\beta^* x \quad (\text{x = } x^*) \qquad (\lambda x.M) \longrightarrow_\beta^* \lambda x.M^* \quad (= (\lambda x.M)^*) \qquad MN \longrightarrow_\beta^* M^*N \longrightarrow_\beta^* M^*N^*$$

$$(\lambda x.L)N \longrightarrow_\beta^* (\lambda x.L^*)N \longrightarrow_\beta^* (\lambda x.L^*)N^* \longrightarrow_\beta L^*\{N^*/x\}.$$

Exercise 6.

$$((\lambda x.\ x\,y)(\lambda a.a))^* = (x\,y)\{\lambda a.a/x\}$$

$$= (\lambda a.a)\,y \longrightarrow_\beta y$$

so not a β-normal form.

Some of the β-redexes in M are still present in M^*.

Exercise 7.

$$\omega_m(M\{N/x\}) = \max\{\omega(R) \mid R \text{ is a } \beta\text{-redex in } M\{N/x\}\}$$
$$= \max\Big(\{\omega(R) \mid R \text{ is a } \beta\text{-redex in } M\}$$
$$\cup \{\omega(R) \mid R \text{ is a } \beta\text{-redex in } N\}$$
$$\cup \{\omega((\lambda x.U)\,a) \mid (\lambda x.U) = N,\ x\,a \in M\}\Big)$$
$$= \max(\omega_m(M), \omega_m(N), size(A)).$$

We can prove this by induction on M:

- $\omega_m(x\{N/x\}) = \omega_m(N) \leq \max(\dots)$
- $\omega_m((\lambda y.M)\{N/x\}) = \omega_m(\lambda y.(M\{N/x\})) = \omega_m(M\{N/x\}) \leq \max(\dots)$
- $\omega_m((x\ M)\{\lambda y.N/x\}) = \omega_m((\lambda y.N)(M\{N/x\}))$
  $\leq \max(size(A), \omega_m(M), \omega_m(N))$

  $(M,N) \neq (x\,U, \lambda y.V)$
- $\omega_m(MM'\{N/x\}) = \omega_m((M\{N/x\})(M'\{N/x\}))$
  $\leq \max(\omega_m(M), \omega_m(M'), \omega_m(N), size(A))$
  $\leq \max(\dots)$

Exercise 8.

By induction on $\Gamma \vdash M : A$,

- case $\Gamma, x:A, \Delta \vdash x : A$, then $\omega_m(x) = 0$.
- case $\Gamma \vdash \lambda x.M : A \to B$, then $\Gamma, x:A \vdash M : B$,

  if $\omega_m(M) = 0$ then $\omega_m(\lambda x.M) = 0$

  if $\omega_m(M^*) < \omega_m(M)$ then

$$\omega_m((\lambda x.M)^*) = \omega_m(\lambda x.M^*)$$
$$= \omega_m(M^*)$$
$$< \omega_m(M)$$
$$= \omega_m(\lambda x.M)$$

- case $\Gamma \vdash MN : A$ then $\Gamma \vdash M : B \to A$ and $\Gamma \vdash N : B$.

if $M = \lambda x.L$, then $(MN)^* = L^*\{N^*/x\}$

and thus

$$\omega_m((MN)^*) = \omega_m(L^*\{N^*/x\})$$
$$\leq \max(\omega_m(L^*), \omega_m(N^*), \text{size}(A))$$

$$\omega_m(MN) = \omega_m((\lambda x\, L)N) \leq \max(\omega_m(L), \omega_m(N), \text{size}(A))$$

and we conclude by ih on L and N.

otherwise $\omega(M^* N^*) \leq \max(\omega_m(M^*), \omega_m(N^*))$

$\omega(M\ N) \leq \max(\omega_m(M\ ), \omega_m(N^*))$

and we conclude.

Tutorial n° 7

Exercise 1.

Let us show the following lemma:

<u>Lemma</u>: if F, F' are CBN-contexts and $R, P \in \mathrm{dom}\,\beta \cup \mathrm{dom}\,\mathcal{C} \cup \mathrm{dom}\,cc$, such that $F[R] = F'[P]$ then $R = P$ and $F = F'$.

<u>Proof</u> by induction on F, F'

- if $F = F' = [\,]$ then $F[R] = R = P = F'[P]$
- if $F[R]\,N = F'[P]\,M$ then we must have $N = M$ and $F[R] = F'[P]$, and we conclude by ih.
- if $F'[R] = R = F[P]\,N$

  then $R = (\lambda x.\, R')\, R''$ or $R = \mathcal{C}\, R'$ or $R = cc\, R'$
  and $P = (\lambda x.\, P')\, P''$ or $P = \mathcal{C}\, P'$ or $P = cc\, P'$
  if $F = [\,]$, then impossible
  if $F = F''[\,]\, M$, then impossible.

For CBV, do it next time.

$\lambda x.\lambda y.\, x$ and $\lambda x.\lambda y.\, y$ are not $\xleftarrow[*]{\mathcal{E}\, cc\, \mathcal{C}}$.

Exercise 3.

Q1. $M\left\{\begin{matrix}[\alpha](LN) \\ \hline [\alpha]\,L\end{matrix}\right\} := \begin{cases} [\alpha](L\{\cdots\}N) & \text{if } M = [\alpha]\,L \\ [\beta]\,L\{\cdots\} & \text{if } M = [\beta]\,L \text{ with } \beta \neq \alpha \\ \mu\alpha.L & \text{if } M = \mu\alpha.\,L \\ \mu\beta.\,L\{\cdots\} & \text{if } M = \mu\beta.\,L \text{ with } \beta \neq \alpha \\ \lambda x.\,L\{\cdots\} & \text{if } M = \lambda x.\,L \\ x & \text{if } M = x \\ L\{\cdots\}\,R\{\cdots\} & \text{if } M = LR \end{cases}$

Q2. For λ-calculus, the substitution lemma is:
if $\Gamma \vdash N : A$ and $\Gamma, x : A \vdash M : B$ then $\Gamma \vdash M\{N/x\} : B$.

Now, if $\Gamma \vdash N : B \mid \Delta$ and $\Gamma \vdash M : C \mid \Delta, \alpha : B \Rightarrow A$ (*)

then $\Gamma \vdash M\left\{\begin{matrix}[\alpha]\,LN \\ \hline [\alpha]\,L\end{matrix}\right\} : C \mid \Delta, \alpha : A$.

We can show it by induction on (*), we only need to consider the yellow cases.

For the case $\mu\,\alpha.\,M$, then we must use rule μ for (*) which is impossible as the premise with two αs, absurd!

For the $[\alpha]M$ case then

$$\Gamma \vdash \underbrace{([\alpha]M)\{\cdots\}}_{[\alpha]\,(M\{\cdots\})\,N} : C \;\Big|\; \Delta, \alpha : A$$